

CSE 432 Robotics

Course Introduction

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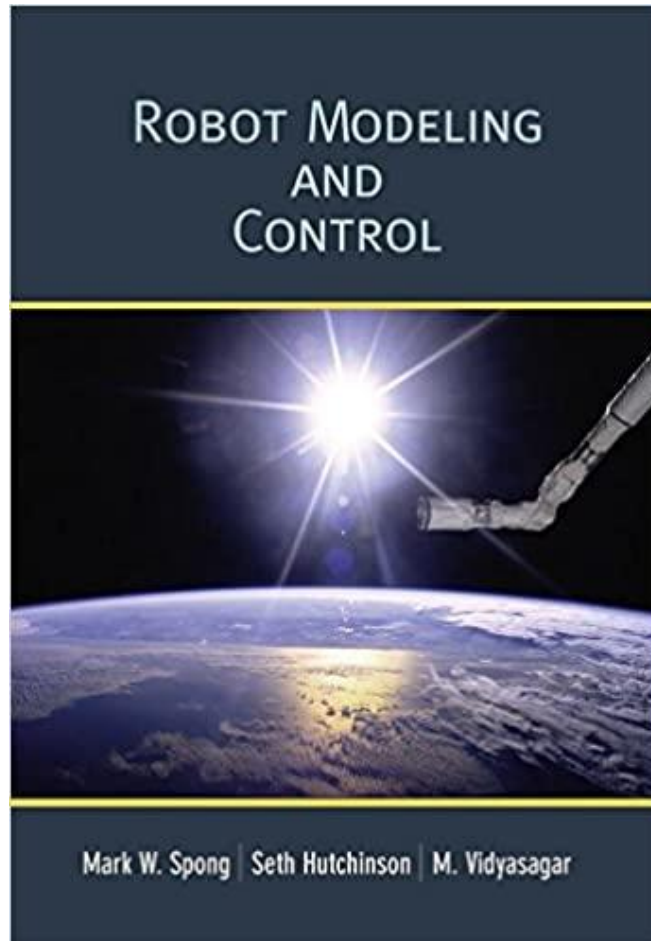
Course Contents

- Introduction to robotics
- Rigid motions and homogeneous transformations
- Forward kinematics
- Inverse kinematics
- Differential kinematics
- Path and trajectory planning



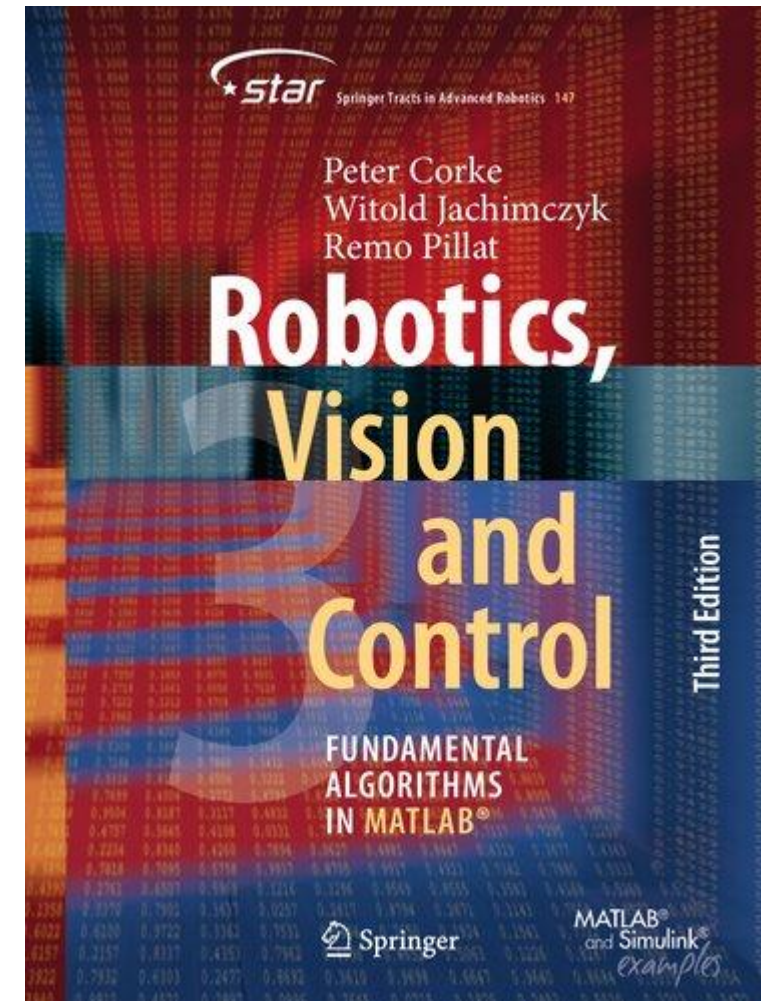
Reference books

- Robot Modeling and Control



Dr. Ahmed Asker

- Robotics, Vision and Control



CSE 432 Robotics



Introduction

- Robotics is a relatively young field of modern technology that crosses traditional engineering boundaries.
- Understanding the complexity of robots and their applications requires knowledge of:
 - Electrical engineering.
 - Mechanical engineering.
 - Systems and industrial engineering.
 - Computer science.
 - Economics, and mathematics.



Introduction

- Robotics encompasses other areas not covered in this course
 - locomotion, including wheeled and legged robots,
 - flying and swimming robots,
 - grasping,
 - artificial intelligence and programming languages.
- This course is concerned with fundamentals of robotics, including **kinematics**, **dynamics**, **motion planning**, and **control**.



Introduction

- Our goal is to provide a complete introduction to the most important concepts in these subjects as applied to industrial robot manipulators.
- This course focus on industrial robot arms operating in structured factory environments.
- Nowadays, robot can do work in proximity of human such as service robots.
- A complete treatment of the discipline of robotics would require several courses.



Introduction

- The term robot was first introduced into our vocabulary by the Czech playwright Karel Capek in his 1920 play Rossum's Universal Robots, the word robota being the Czech word for work.
- Since then, the term has been applied to a great variety of mechanical devices, such as tele-operators, underwater vehicles, autonomous land rovers, etc.



Introduction

- Virtually anything that operates with some degree of autonomy, usually under computer control, has at some point been called a robot.
- In this course the term robot will mean a **computer controlled industrial manipulator**.



What is a Robot?

- Robotics
 - Scientific area that studies the link between Perception and Action
- Robot Arms
 - Device able to perform activities as a human
 - Programmable manipulator able to execute multiple operations, following programmed paths to fulfill a large variety of tasks.
- Mobile Robot / Mobile Platform
 - Platform with a large mobility within its environment (air, land, underwater)
 - A system with the following functional characteristics:
 - Mobility: total mobility relative to the environment
 - A certain level of autonomy: limited human interaction
 - Perception ability: sensing and reacting in the environment



What is a Robot?

- Is a washing machine a robot? Most people would call it an appliance instead, but it does have sensing, actuation and processing.
- A possible distinction between appliance and robot whether the workspace is physically inside or outside the device.
- The cognitive ability required of a robot is much higher: the outside world is complex, and harder to understand and control.
- What about a modern car? Or smartphone? Are they becoming robots?





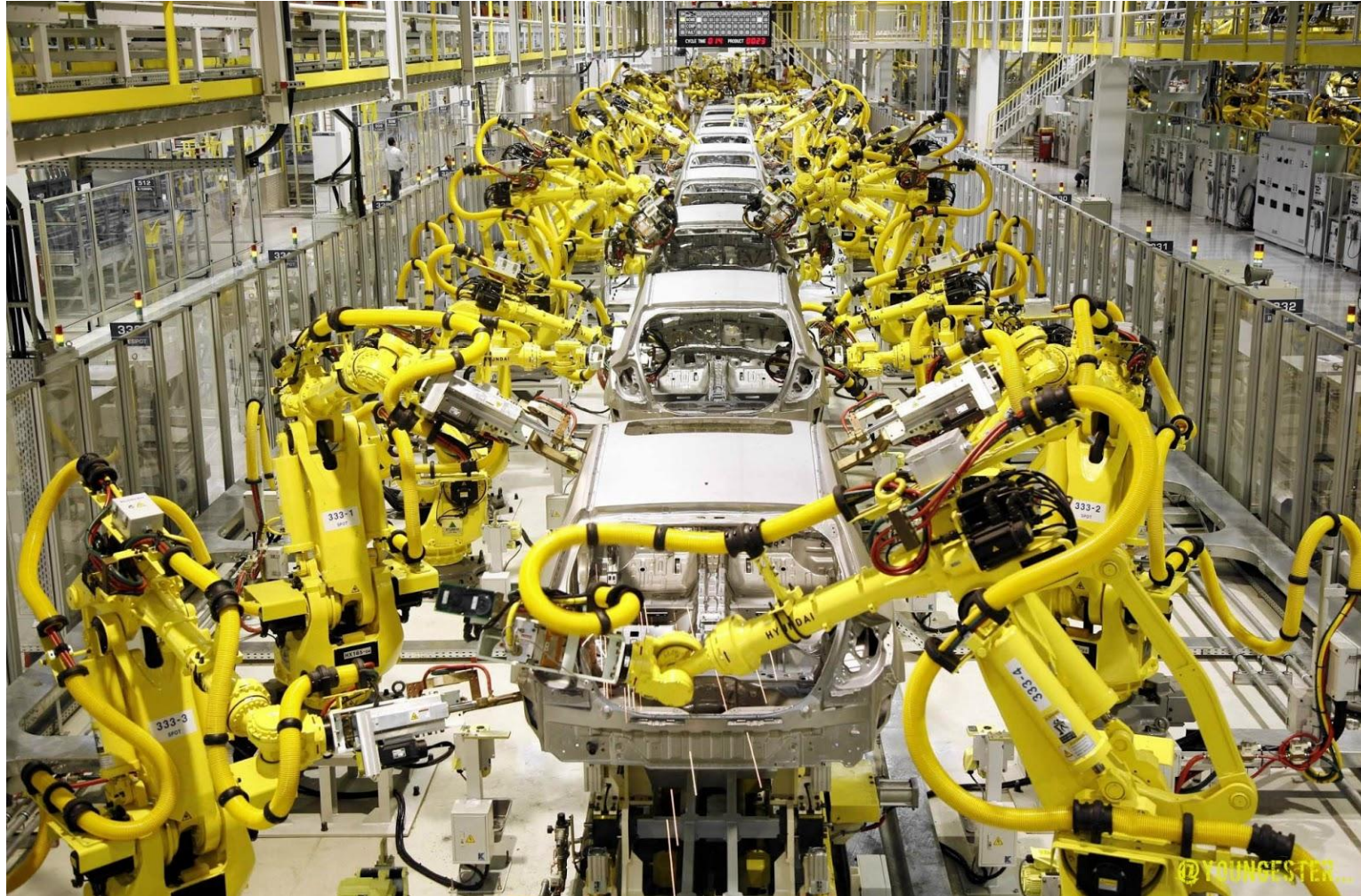
Robots in automation

- An official definition of such a robot comes from the Robot Institute of America (RIA):

“A robot is a **reprogrammable multifunctional** manipulator designed to move material, parts, tools, or specialized devices through **variable programmed motions** for the performance of a **variety of tasks**”.



Industrial Robots



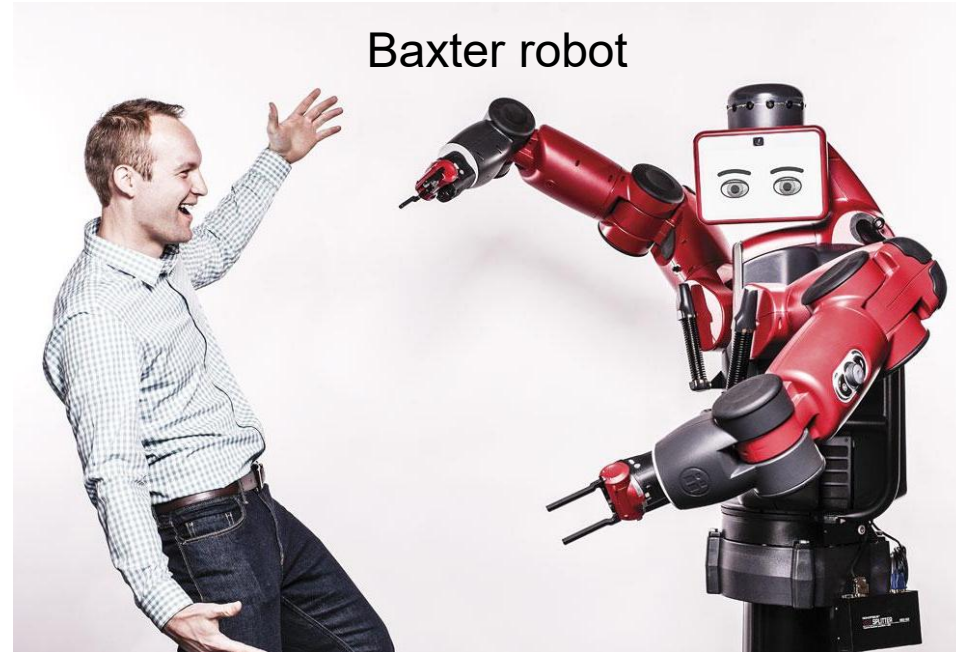


Field Robots





Social Robots



**Kaspar the
friendly
robot**



Autonomous robots

Quad-Rotor



Autonomous under-water vehicle



Autonomous car!



Spot mini



Surgical Robot



da Vinci Surgical System



MATHEMATICAL MODELING OF ROBOTS

- While robots are themselves mechanical systems, in this course we will be primarily concerned with developing and manipulating mathematical models for robots.
- We will develop methods to represent basic geometric aspects of robotic manipulation, dynamic aspects of manipulation, and the various sensors available in modern robotic systems.

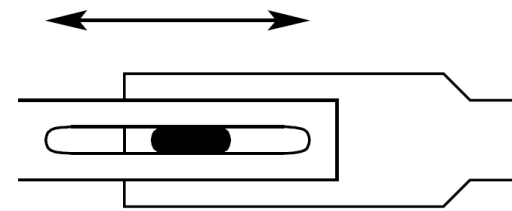
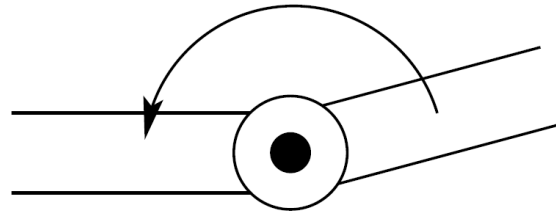


Type of Joints

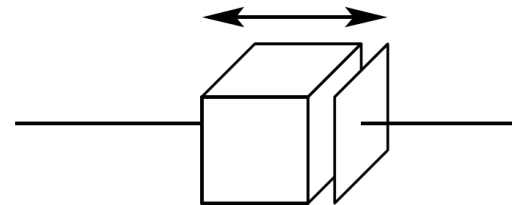
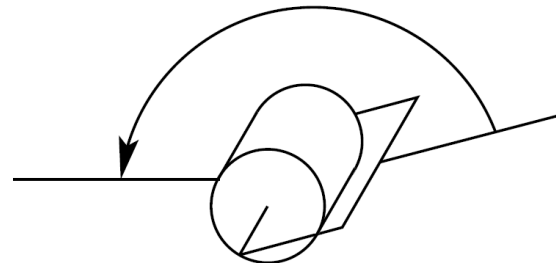
Revolute

Prismatic

2D



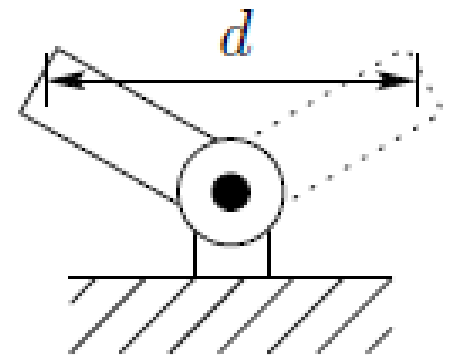
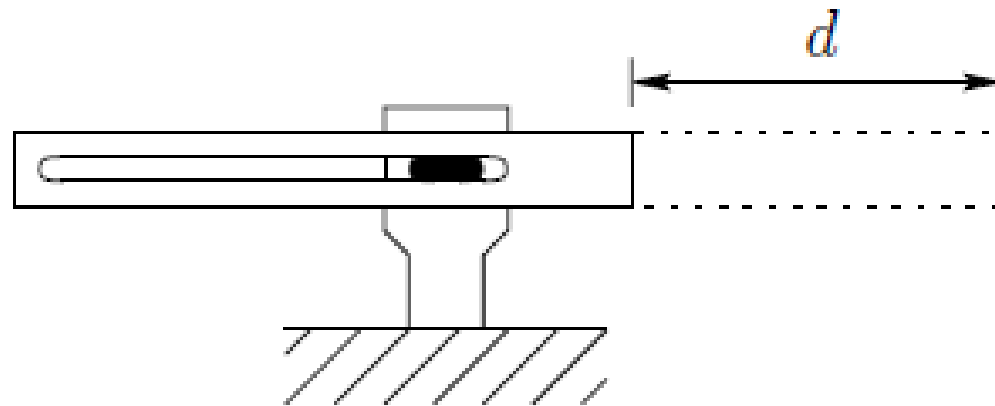
3D





Linear vs Rotational Link

- Prismatic joints, typically have higher resolution than revolute joint.
- Rotational axes usually result in a large amount of kinematic and dynamic coupling among the links with a resultant accumulation of errors and a more difficult control problem.
- Revolute joint increases dexterity and compactness of the manipulator.





The Workspace

- The workspace of a manipulator is the total volume swept out by the end-effector as the manipulator executes all possible motions.
- The workspace is constrained by the geometry of the manipulator as well as mechanical constraints on the joints.
- For example, a revolute joint may be limited to less than a full 360 of motion.



The Workspace

- The workspace is often broken down into a reachable workspace and a dexterous workspace.
- **Reachable workspace** is the entire set of points reachable by the manipulator,
- **Dexterous workspace** consists of those points that the manipulator can reach with an arbitrary orientation of the end-effector.
- Obviously, the dexterous workspace is a subset of the reachable workspace.



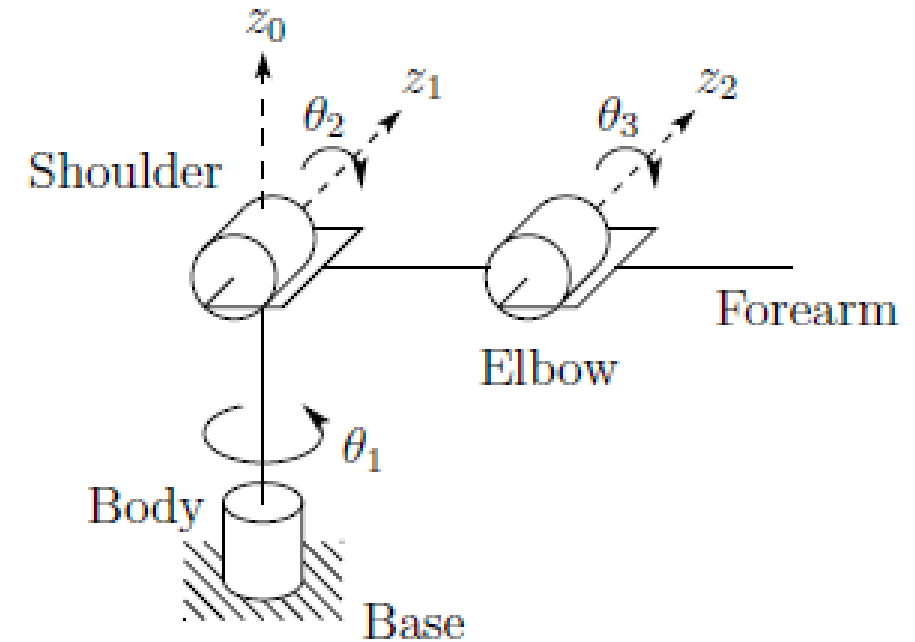
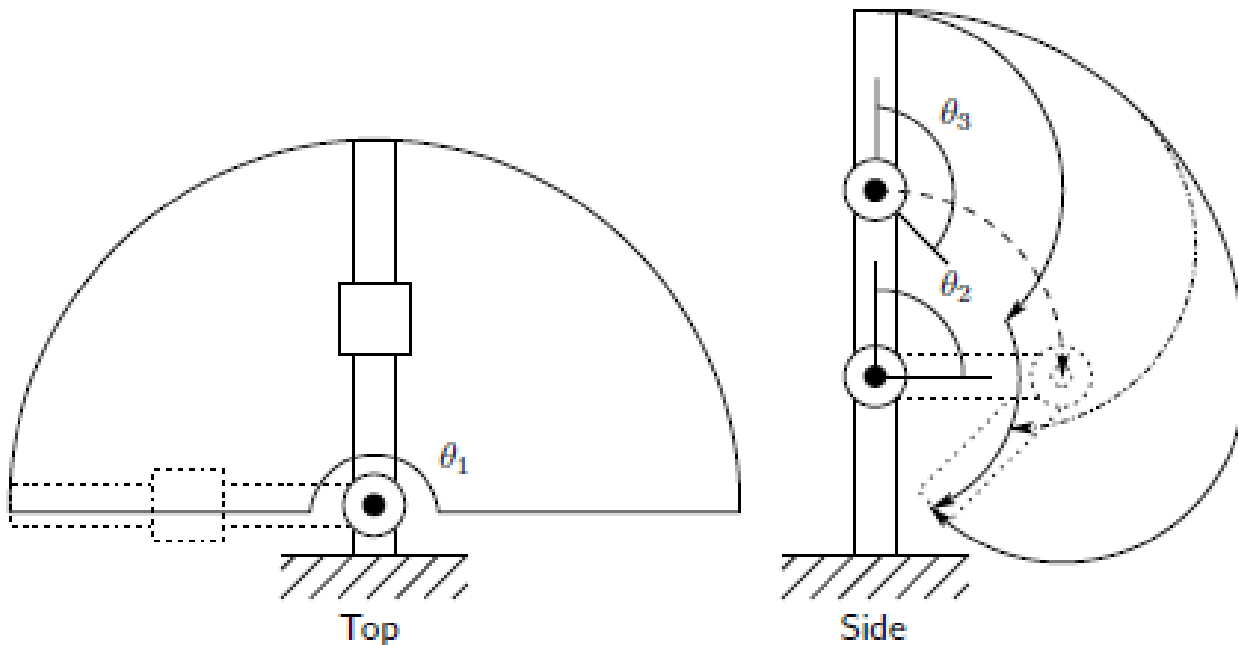
Articulated manipulator (RRR)





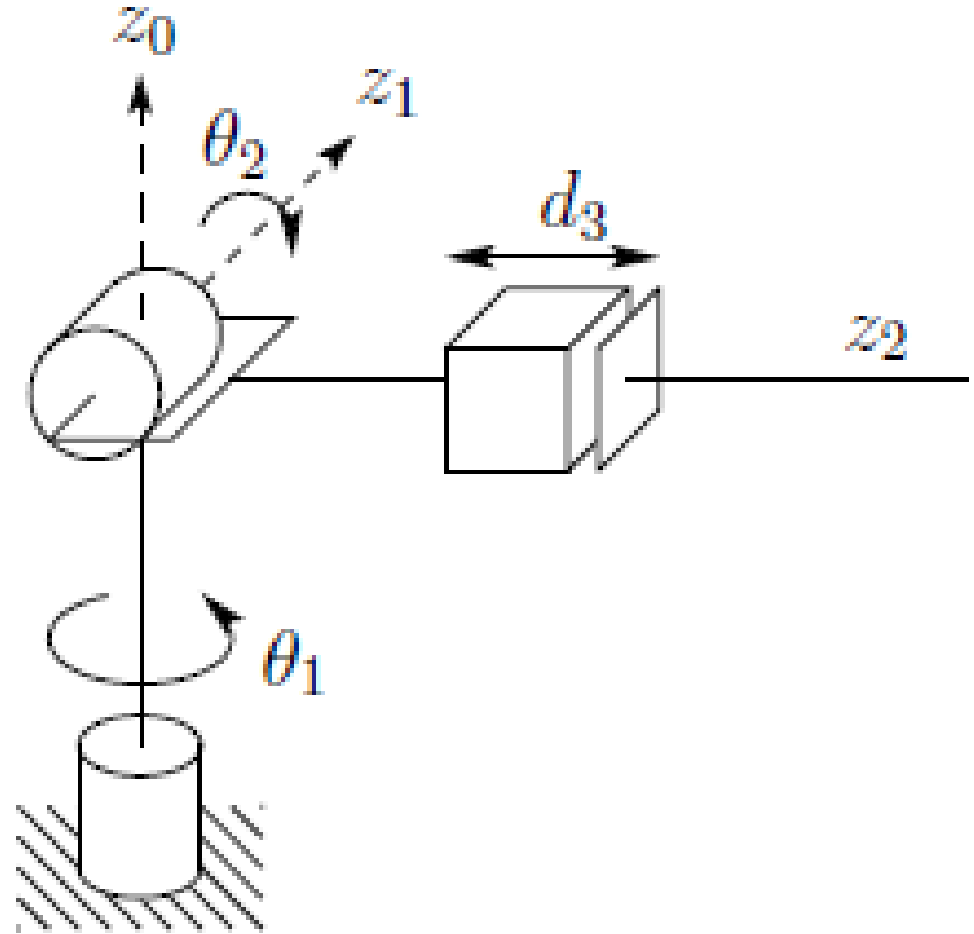
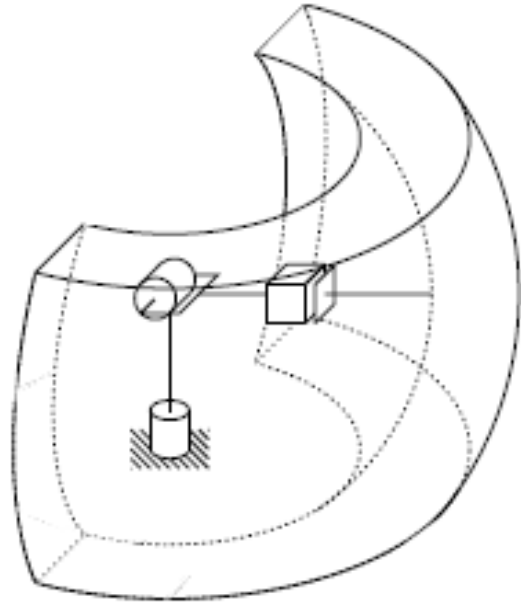
Articulated manipulator (RRR)

- In these arrangements joint axis z_2 is parallel to z_1 and both z_1 and z_2 are perpendicular to z_0 .
- This kind of manipulator is known as an elbow manipulator.



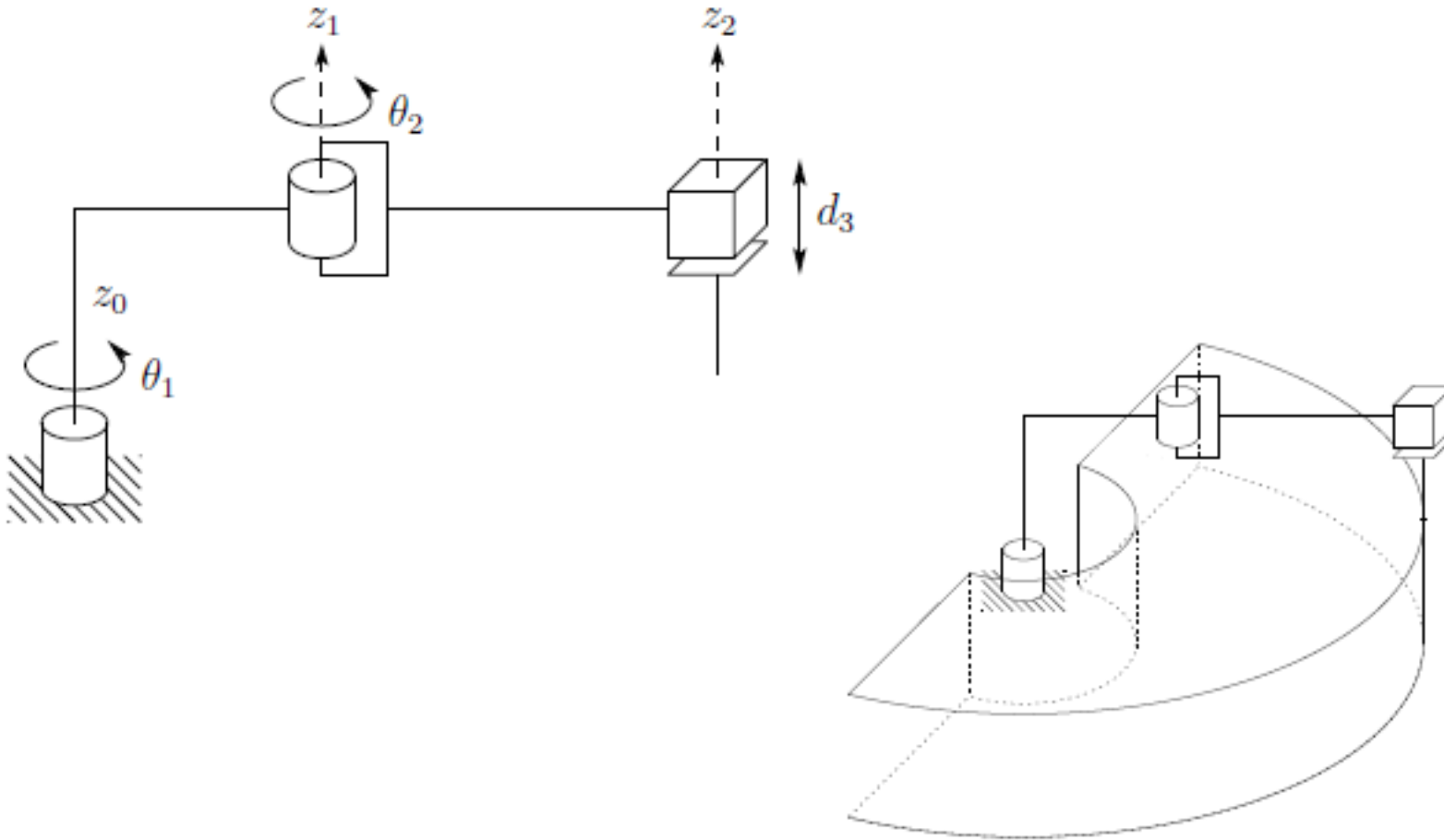


Spherical Manipulator (RRP)



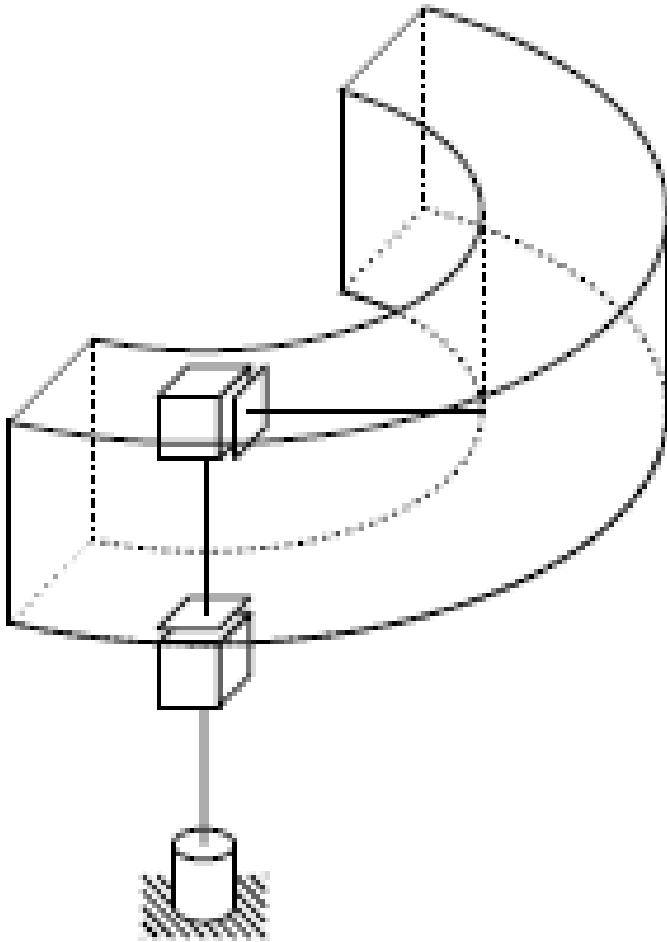


SCARA Manipulator (RRP)

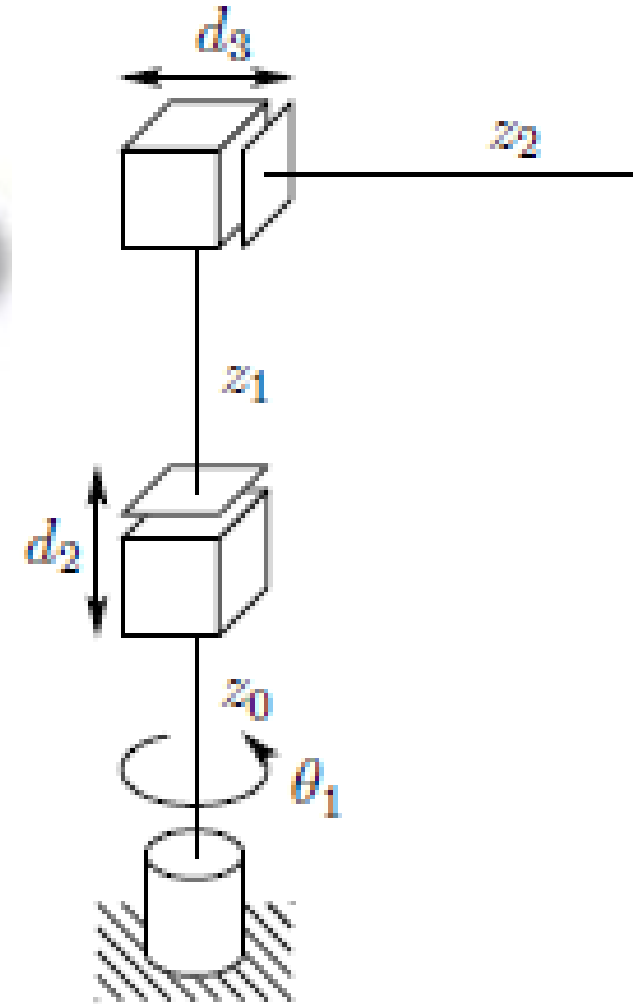




Cylindrical Manipulator (RPP)

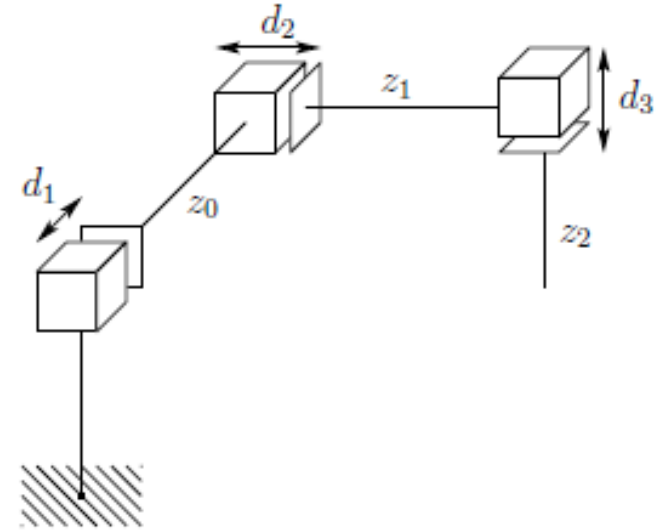
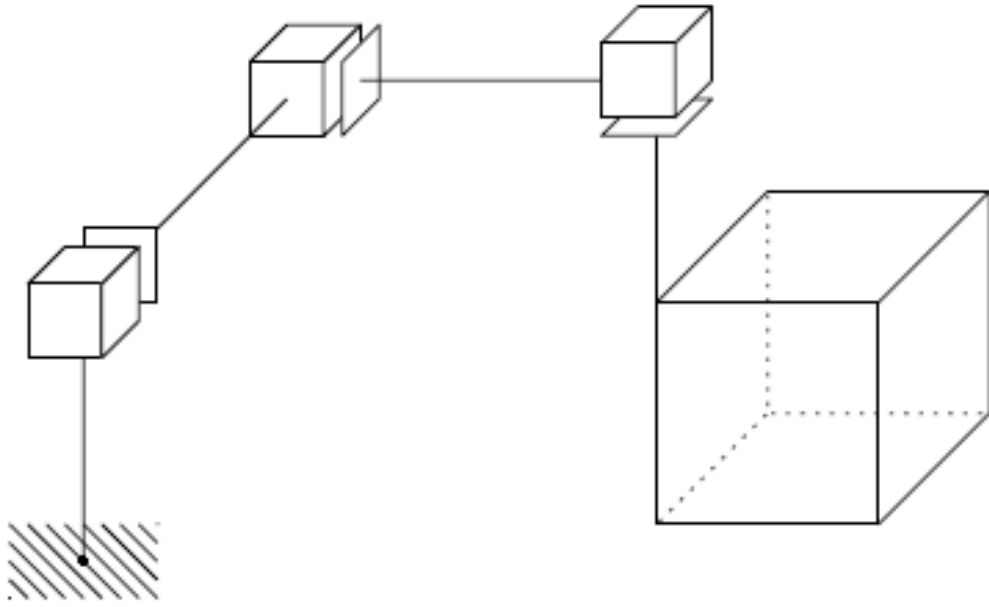


Seiko RT3300



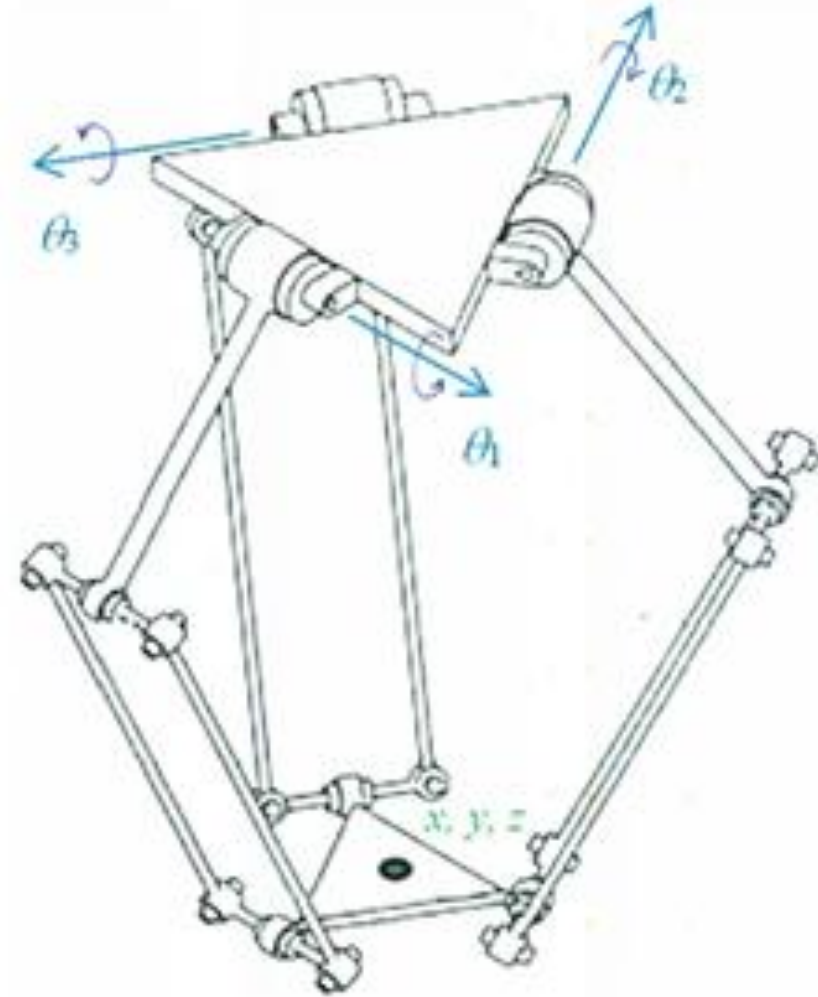


Cartesian manipulator (PPP)





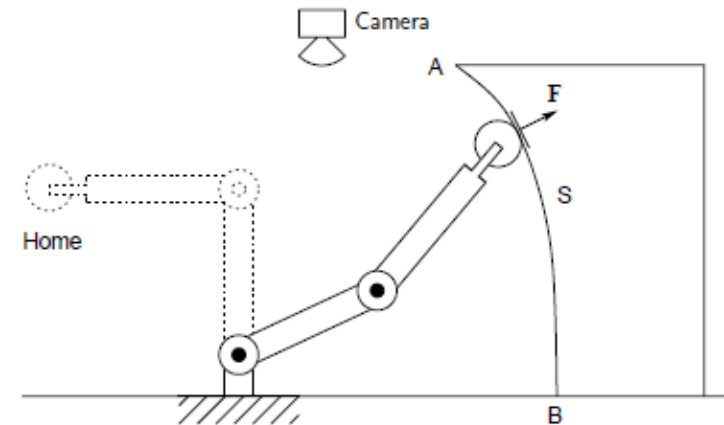
Parallel Manipulator





Example of robotics problem

- The first problem encountered is to describe both the position of the tool and the locations A and B with respect to a common coordinate system.
- Typically, the manipulator will be able to sense its own position in some manner using internal sensors (position encoders located at joints 1 and 2) that can measure directly the joint angles 1 and 2.
- We also need therefore to express the positions A and B in terms of these joint angles.





Forward Kinematics

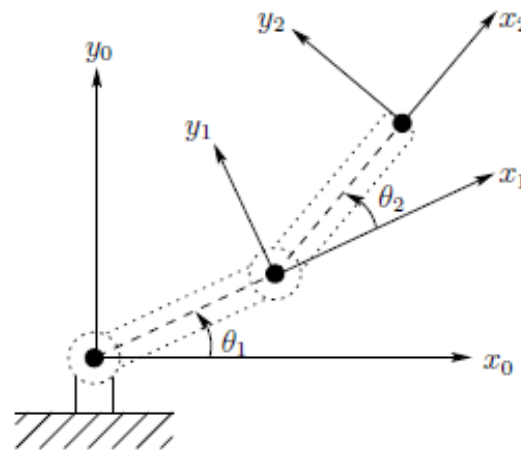
- The coordinates (x, y) of the tool are expressed in this coordinate frame as

$$x = x_2 = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2)$$

$$y = y_2 = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2)$$

- The orientation of the tool frame relative to the base frame is given by the direction cosines of the x_2 and y_2 axes relative to the x_0 and y_0 axes

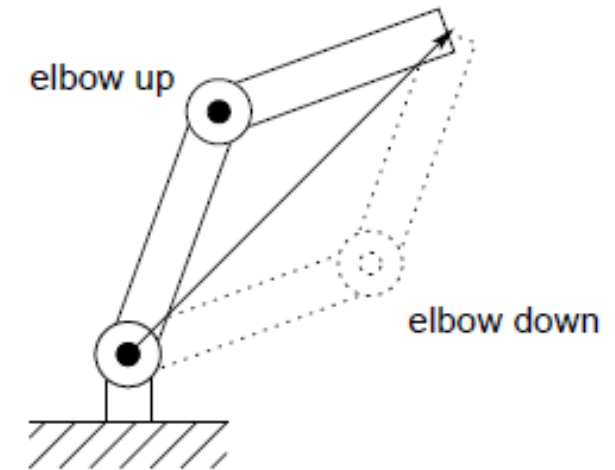
$$\begin{bmatrix} x_2 \cdot x_0 & y_2 \cdot x_0 \\ x_2 \cdot y_0 & y_2 \cdot y_0 \end{bmatrix} = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{bmatrix}$$





Inverse Kinematics

- In order to command the robot to move to location A we need the joint variables θ_1, θ_2 in terms of the x and y coordinates of A.
- If the given (x, y) coordinates are within the manipulator's reach there may be two solutions, the so-called
 - **elbow up** configurations
 - **elbow down** configurations.



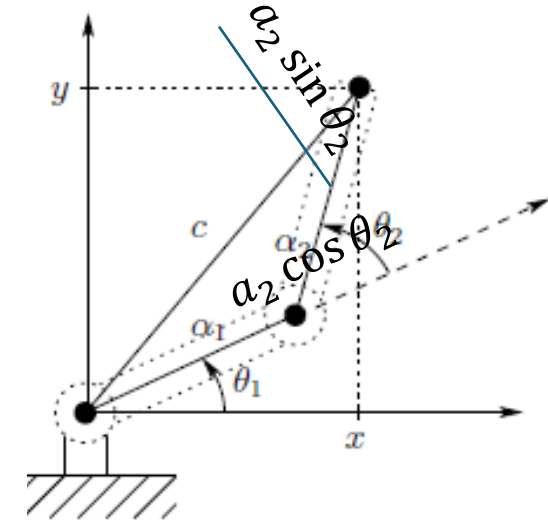
- There may be exactly one solution if the manipulator must be fully extended to reach the point.



Inverse Kinematics

- Using the Law of Cosines, the angle θ_2 is given by

$$\cos \theta_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2} := D$$
$$\sin \theta_2 = \pm \sqrt{1 - D^2}$$
$$\theta_2 = \tan^{-1} \frac{\pm \sqrt{1 - D^2}}{D}$$



- The advantage of this latter approach is that both the elbow-up and elbow down solutions are recovered by choosing the positive and negative signs

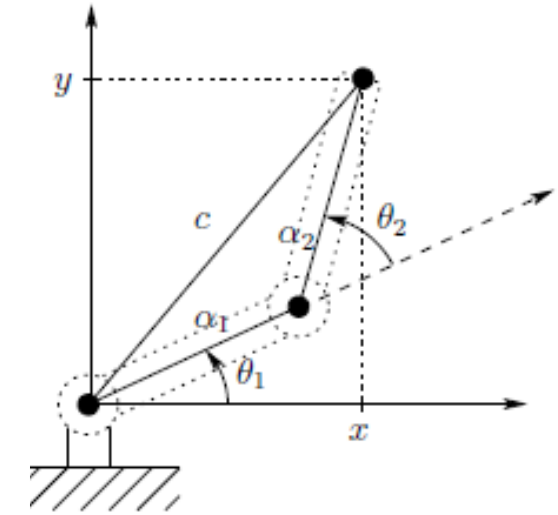
$$\theta_1 = \tan^{-1} \left(\frac{y}{x} \right) - \tan^{-1} \left(\frac{a_2 \sin \theta_2}{a_1 + a_2 \cos \theta_2} \right)$$

- Notice that, θ_1 depends on θ_2 . Thus, different value for θ_1 , is obtained depending on which solution is chosen for θ_2 .



Velocity Kinematics

- The position of the end-effector (x, y)
$$x = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2)$$
$$y = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2)$$
- The end-effector velocity can be obtained by differentiating the position equation



$$\dot{x} = -a_1 \sin \theta_1 \dot{\theta}_1 - a_2 \sin(\theta_1 + \theta_2) (\dot{\theta}_1 + \dot{\theta}_2)$$
$$\dot{y} = a_1 \cos \theta_1 \dot{\theta}_1 + a_2 \cos(\theta_1 + \theta_2) (\dot{\theta}_1 + \dot{\theta}_2)$$

- Using the vector notation $\mathbf{x} = [x, y]^T$, $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$

$$\dot{\mathbf{x}} = \mathbf{J} \dot{\boldsymbol{\theta}}$$

$$\mathbf{J} = \begin{bmatrix} -a_1 \sin \theta_1 \dot{\theta}_1 - a_2 \sin(\theta_1 + \theta_2) & -a_2 \sin(\theta_1 + \theta_2) \\ a_1 \cos \theta_1 \dot{\theta}_1 + a_2 \cos(\theta_1 + \theta_2) & a_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$



Velocity Kinematics

- The joint velocities are found from the end-effector velocities via the inverse Jacobian

$$\dot{\theta} = J^{-1} \dot{x}$$

where

$$J^{-1} = \frac{1}{a_1 a_2 s_2} \begin{bmatrix} a_2 c_{12} & a_2 s_{12} \\ -a_1 c_1 - a_2 c_{12} & -a_1 s_1 - a_2 s_{12} \end{bmatrix}$$

- The Jacobian does not have an inverse, $\theta_2 = 0$ or π , in which case the manipulator is said to be in a singular configuration.



Path Planning and Trajectory Generation

- The robot control problem is typically decomposed hierarchically into three tasks:
 - Path planning
 - Trajectory generation
 - Trajectory tracking.
- The **path planning** problem is to determine a path in task space (or configuration space) to move the robot to a goal position while avoiding collisions with objects in its workspace.



Path Planning and Trajectory Generation

- These paths encode position and orientation information without timing considerations, i.e. without considering velocities and accelerations along the planned paths.
- The **trajectory generation** problem, is to generate reference trajectories that determine the time history of the manipulator along a given path or between initial and final configurations.



Dynamics

- A robot manipulator is primarily a positioning device.
- To control the position, we must know the dynamic properties of the manipulator to know how much force to exert on it to cause it to move:
 - too little force and the manipulator is slow to react;
 - too much force and the arm may crash into objects or oscillate about its desired position.
- Deriving the dynamic equations of motion for robots is not a simple task due to the large number of degrees of freedom and nonlinearities present in the system.



Position Control

- The motion control problem consists of
 - **Tracking**
 - **Disturbance Rejection**
- This aims to determine the control inputs necessary to follow, or **track**, a desired trajectory that has been planned for the manipulator, while simultaneously **rejecting** disturbances due to un-modeled dynamic effects such as friction and noise.



Course Projects

- 5 points for getting the certificate + Documenting your online training (<https://academy.universal-robots.com/free-e-learning/ur2030-e-learning/>).
- Group project: Modelling and control of 6 DOF robotic arm using MATLAB Robotics System Toolbox (10 points).
- Group project presentation (5 points).
- A useful repository to find URDF files for different robots:
https://github.com/Daniella1/urdf_files_dataset/blob/main/README.md



Useful Trigonometric Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

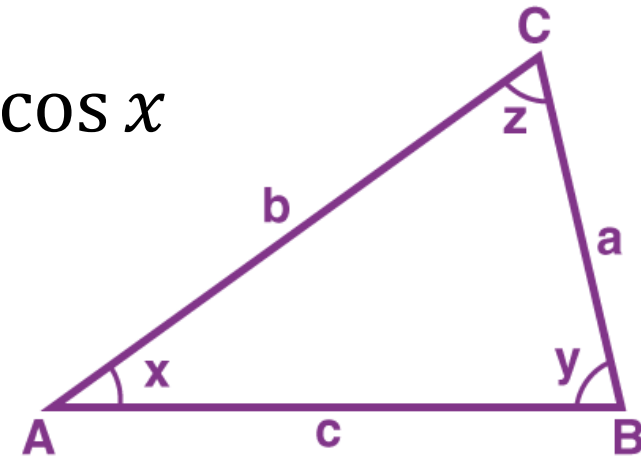
$$\sin \theta = \pm \sqrt{1 - \cos^2 \theta}$$

$$\cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

$$\sin(\theta_1 \pm \theta_2) = \sin \theta_1 \cos \theta_2 \pm \cos \theta_1 \sin \theta_2$$

$$\cos(\theta_1 \pm \theta_2) = \cos \theta_1 \cos \theta_2 \mp \sin \theta_1 \sin \theta_2$$

$$a^2 = b^2 + c^2 - 2bc \cos x$$





Any Questions